CLASS 8 · CBSE · GANITA PRAKASH PART 1 · SHORT NOTES

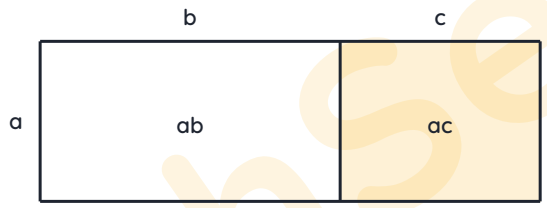
Chapter 6: We Distribute, Yet Things Multiply

Topic-wise short notes with solved examples

The distributive property and what it leads to: multiplying two brackets, the identities for the square of a sum, the square of a difference and the difference of two squares, quick multiplication methods, and describing patterns with algebra.

1 The distributive property

- Multiplication distributes over addition: $a(b+c) = ab + ac$ and $(a+b)c = ac + bc$.
- It also works for subtraction, $a(b-c) = ab - ac$, and for any number of terms inside the bracket.
- It holds for all integers and fractions, positive or negative.



A rectangle a by (b + c) has area ab + ac

- An **identity** is an equality between two expressions that is true for every value of the letters. Example: $a(b+8) = ab + 8a$.
- Only **like terms** (same letters with the same powers) can be added: $a^2b + 2a^2b = 3a^2b$, but $\frac{3}{2}a^2$ and $\frac{3}{10}a$ cannot be combined.

Example 1.1 Expand $\frac{3a}{2}\left(a - b + \frac{1}{5}\right)$.

SOLUTION

Multiply each term by $\frac{3a}{2}$.

$\frac{3}{2}a^2 - \frac{3}{2}ab + \frac{3}{10}a$

Example 1.2 By how much does the product 23×27 increase if 27 is increased by 1?

SOLUTION

$23(27+1) = 23 \times 27 + 23$

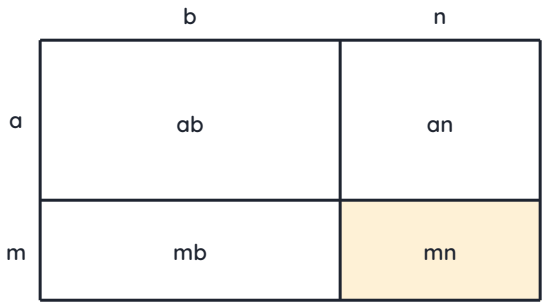
By 23

2 Multiplying two brackets

- Multiply each term of the first bracket by each term of the second, then add.

$$(a+m)(b+n) = ab + an + mb + mn$$

- With signs, use the usual rules for integers: $(a+u)(b-v) = ab - av + ub - uv$, $(a-u)(b+v) = ab + av - ub - uv$ and $(a-u)(b-v) = ab - av - ub + uv$.



(a + m)(b + n) is the sum of four areas

- **How a product changes**: if both numbers go up by 1, $(a+1)(b+1) = ab + a + b + 1$, so the product rises by $a + b + 1$. If one goes up by 1 and the other down by 1, $(a+1)(b-1) = ab + b - a - 1$.

Example 2.1 Expand $(3-x)(x-6)$.

SOLUTION

$3x - 18 - x^2 + 6x$

$-x^2 + 9x - 18$

Example 2.2 Expand $(4y+7)(y+11z-3)$.

SOLUTION

$4y^2 + 44yz - 12y + 7y + 77z - 21$

$4y^2 + 44yz - 5y + 77z - 21$

Example 2.3 The product of two numbers stays the same when one is increased by 2 and the other is decreased by 4. How are the numbers related?

SOLUTION

$(a+2)(b-4) = ab$ gives $ab - 4a + 2b - 8 = ab$, so $2b = 4a + 8$.

$b = 2a + 4$, **for example 1 and 6, or 3 and 10**

Example 2.4 Which is larger: 25×75 or 26×74?

SOLUTION

$26 \times 74 = (25+1)(75-1) = 25 \times 75 + 75 - 25 - 1 = 25 \times 75 + 49$

26×74 **is larger, by 49**

Example 2.5 Expand $(a+b)(a^2+2ab+b^2)$.

SOLUTION

$a^3 + 2a^2b + ab^2 + a^2b + 2ab^2 + b^3$

$a^3 + 3a^2b + 3ab^2 + b^3$

3 Quick multiplication

- **By 11**: $N \times 11 = N \times 10 + N$. Writing from the right: the units digit, then the sum of each pair of neighbouring digits (carrying when needed), then the first digit.
- **By 101**: $N \times 101 = N \times 100 + N$. **By 1001**: $N \times 1000 + N$.
- **By 99, 999**: $N \times 99 = N \times 100 - N$ and $N \times 999 = N \times 1000 - N$.
- Brahmagupta (628 CE) stated the distributive property as "multiplication by parts" and used it for fast multiplication.

Example 3.1 Find 3874×11 in one line.

SOLUTION

From the right: 4; $7+4=11$, write 1 carry 1; $8+7+1=16$, write 6 carry 1; $3+8+1=12$, write 2 carry 1; $3+1=4$.

42614

Example 3.2 Find 9734×99.

SOLUTION

$9734 \times 100 - 9734 = 973400 - 9734$

963666

Example 3.3 Find 265×1001.

SOLUTION

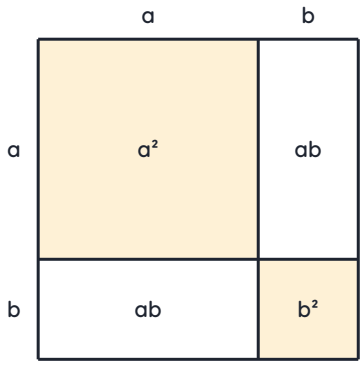
$265000 + 265$

265265

4 Square of a sum

$$(a+b)^2 = a^2 + 2ab + b^2$$

- Reason: $(a+b)(a+b) = a^2 + ab + ba + b^2$.
- Picture: a square of side $a+b$ splits into squares of area a^2 and b^2 and two rectangles of area ab each.



$(a + b)^2 = a^2 + 2ab + b^2$

- $(a+b)^2$ is not $a^2 + b^2$. The middle term $2ab$ must be included.

Example 4.1 Find 65^2 and 104^2.

SOLUTION

$65^2 = (60+5)^2 = 3600 + 600 + 25 = 4225$

$104^2 = (100+4)^2 = 10000 + 800 + 16 = 10816$

4225 and 10816

Example 4.2 Expand $(6x+5)^2$.

SOLUTION

$(6x)^2 + 2(6x)(5) + 5^2$

$36x^2 + 60x + 25$

5 Square of a difference

$$(a-b)^2 = a^2 - 2ab + b^2$$

- It follows from the first identity by writing $a-b$ as $a+(-b)$.
- $(a-b)^2 = (b-a)^2$, since squaring removes the sign.

Example 5.1 Find 99^2 and 58^2.

SOLUTION

$99^2 = (100-1)^2 = 10000 - 200 + 1 = 9801$

$58^2 = (60-2)^2 = 3600 - 240 + 4 = 3364$

9801 and 3364

Example 5.2 Expand $\left(7y - \frac{3}{4}\right)^2$.

SOLUTION

$(7y)^2 - 2(7y)\left(\frac{3}{4}\right) + \left(\frac{3}{4}\right)^2$

$49y^2 - \frac{21}{2}y + \frac{9}{16}$

6 Difference of two squares

$$(a+b)(a-b) = a^2 - b^2$$

- Reason: $a^2 - ab + ba - b^2$, and the middle terms cancel.
- It turns a product of two numbers equally far from a round number into a subtraction of squares.
- **Śrīdharācārya's method for squares** (750 CE): $a^2 = (a+b)(a-b) + b^2$. Choose b so that $a+b$ or $a-b$ is a round number.

Example 6.1 Find 98×102 and 45×55.

SOLUTION

$98 \times 102 = (100-2)(100+2) = 10000 - 4$

$45 \times 55 = (50-5)(50+5) = 2500 - 25$

9996 and 2475

Example 6.2 Find 197^2 by Śrīdharācārya's method.

SOLUTION

$197^2 = (197+3)(197-3) + 3^2 = 200 \times 194 + 9$

38809

Example 6.3 Write 100 as a difference of two squares.

SOLUTION

$100 = 50 \times 2 = (26+24)(26-24)$

$26^2 - 24^2$

7 Patterns explained by identities

- Adding the first two identities: $(a+b)^2+(a-b)^2=2(a^2+b^2)$. So twice a sum of two squares is again a sum of two squares, for example $2(5^2+3^2)=8^2+2^2$.
- Subtracting them: $(a+b)^2-(a-b)^2=4ab$.
- Three consecutive numbers: $n^2-(n-1)(n+1)=1$. The square of the middle one is 1 more than the product of the other two.
- $(a-b)(a+b)=a^2-b^2$, $(a-b)(a^2+ab+b^2)=a^3-b^3$, $(a-b)(a^3+a^2b+ab^2+b^3)=a^4-b^4$.

Example 7.1 In a calendar, take any 2×2 block of dates. Show that the two diagonal products always differ by 7.

SOLUTION

Call the dates a, $a+1$, $a+7$, $a+8$.

$(a+1)(a+7)-a(a+8)=a^2+8a+7-a^2-8a$

7, whatever the value of a

Example 7.2 Show that the square of an odd number is 1 more than a multiple of 8.

SOLUTION

$(2n+1)^2=4n^2+4n+1=4n(n+1)+1$

$n(n+1)$ is a product of consecutive numbers, so it is even. Hence $4n(n+1)$ is a multiple of 8.

Proved

8 Many ways to count

- The same pattern can be counted in different ways. Each way gives a different-looking expression, and simplifying shows they are equal.

Example 8.1 A pattern has k rows of $k+2$ dots at step k. Show that $k(k+2)$, $(k+1)^2-1$ and k^2+2k all describe it, and find the number of dots at step 15.

SOLUTION

$k(k+2)=k^2+2k$ and $(k+1)^2-1=k^2+2k+1-1=k^2+2k$

All are equal. At step 15: $225+30=255$ dots.

Example 8.2 Four rectangles, each m by n with $n>m$, are arranged around a square hole to form a large square of side $m+n$. Find the area of the hole in two ways.

SOLUTION

Way 1: large square minus four rectangles, $(m+n)^2-4mn$.

Way 2: the hole is a square of side $n-m$, area $(n-m)^2$.

Both equal $m^2-2mn+n^2$

9 Common mistakes

Wrong	Right
$(5m+6n)^2 = 25m^2 + 36n^2$	$25m^2 + 60mn + 36n^2$
$2(x-1)+3(x+4) = 2x-1+3x+4$	$2x-2+3x+12 = 5x+10$
$-3p(-5p+2q) = -3p+5p-2q$	$15p^2 - 6pq$
$(a+2)(b+4) = ab+8$	$ab+4a+2b+8$
$5w^2 + 6w = 11w^2$	Unlike terms; cannot be combined
$3a(2b \times 3c) = 6ab \times 9ac$	$18abc$ (no distribution over multiplication)
$y+2(y+2) = (y+2)^2$	$y+2y+4 = 3y+4$

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